Comsats University Islamabad, Lahore Campus

Terminal Examination – FALL 2018

Course Title	Information Security	Course Code		CSC432	Credit Hours		3(2,1)
Instructor/s	Dr. Adnan Ahmad, Dr. Muhammad Sharjeel	Program Name		BSSE			
Semester	8th	Batch	FA12	Section	A,B	Date	Dec 20, 2016
Time Allowed	3 hours	Maximum Marks					
Student Name:		Registration No.					

Important Instructions/Guidelines:

1. Avoid writing long stories.
2. Use of S-Box, Lookup table cheat sheet is allowed
3. Exchange of calculator is not allowed
4. Match of solution from a certain threshold will result in failure of entire course!
5. Return the question paper along with the answer sheet. **Missing paper will result in elimination of 5 marks**

Question no. 1 **[10+5+5]; PLO-3; CLO :<3>; Bloom Taxonomy Level: <Analyzing>**

1. Given the plaintext {**00010203040506070809 0A0B0C0D0E0F**} and the key {01010101010101010101010101010101}: a. Show the original contents of State, displayed as a 4 × 4 matrix.
b. Show the value of State after initial AddRoundKey.
c. Show the value of State after SubBytes.
d. Show the value of State after ShiftRows.
e. Show the value of State after MixColumns.

2. This problem explores the use of a one-time pad version of the Vigenère cipher. In this scheme, the key is a stream of random numbers between 0 and 26. For example, if the key is 3 19 5...,then the first letter of plaintext is encrypted with a shift of 3 letters, the second with a shift of 19 letters, the third with a shift of 5 letters, and so on.
a. Encrypt the plaintext "sendmoremoney" with the key stream 9 0 1 7 23 15 21 14 11 11 2 8 9
b. Using the cipher text produced in part (a), find a key so that the cipher text decrypts to the plaintext "cashnotneeded".

3. For each of the following equations, find an integer that satisfies the equation.
a. 5x = 4 mod 3
b. 9x = 8 mod 7
c. -5 mod 3

Question no. 2 **[3+6+3+3]; PLO-4; CLO :<3,4>; Bloom Taxonomy Level: <Applying>**

1. Alice in realm 1 wants to login to a desktop in realm 2 using pre-registered credentials. Explain how Kerberos is involved in this entire process.

2. Users A and B use the Diffie-Hellman key exchange technique with a common prime q = 71 and a primitive root a = 7
a. If user A has private key. $X_A = 5$
b. If user B has private key, what is A's public key X_B = 12 YA, what is B's public key? Y_B?
c. What is the shared secret key?

3. State the value of the length field in SHA-512 if the length of the message is
a. 1919 bits
b. 1920 bits
c. 1921 bits

4. Alice has sent a digital document on public holiday to Bob claiming the notification. How does Bob system will verify that the document is authentic?

<u>Question no</u>. 3 **[6+2+2]; PLO-3; CLO :<1,4>; Bloom Taxonomy Level: <Applying>**

Prior to the discovery of any specific public-key schemes, such as RSA, an existence proof was developed whose purpose was to demonstrate that public-key encryption is possible in theory.

Consider the functions f1(x1) = z1; f2(x2, y2) = z2; f3(x3, y3) = z3, where all values are integers with $1 \leq xi, yi, zi \leq N$.

Function f1 can be represented by a vector M1 of length N, in which the kth entry is the value of f1(k). Similarly,f2 and f3 can be represented by N×N matrices M2 and M3.The intent is to represent the encryption/decryption process by table lookups for tables with very large values of N. Such tables would be impractically huge but could be constructed in principle. The scheme works as follows: Construct M1 with a random permutation of all integers between 1 and N; that is, each integer appears exactly once in M1. Construct M2 so that each row contains a random permutation of the first N integers. Finally, fill in M3 to satisfy the following condition:

To summarize,

1. M1 takes an input k and produces an output x.

2. M2 takes inputs x and p giving output z.

3. M3 takes inputs z and k and produces p.

The three tables, once constructed, are made public.

a. It should be clear that it is possible to construct M3 to satisfy the preceding condition. As an example, fill in M3 for the following simple case:

M1 =

5
4
2
3
1

M2 =

5	2	3	4	1
4	2	5	1	3
1	3	2	4	5
3	1	4	2	5
2	5	3	4	1

M3 =

b. Describe the use of this set of tables to perform encryption and decryption between two users.

c. Argue that this is a secure scheme.

<u>Question no</u>. 4 **[1+1+1+1+1]; PLO-2; CLO :<1>; Bloom Taxonomy Level: <Remembering>**

A) For each of the following assets, assign a low, moderate, or high impact level for the loss of confidentiality, availability, and integrity, respectively. Justify your answers.

a. An organization managing public information on its Web server.
b. A law enforcement organization managing extremely sensitive investigative information.
c. A financial organization managing routine administrative information (not privacy related information).
d. An information system used for large acquisitions in a contracting organization contains both sensitive, pre-solicitation phase contract information and routine administrative information. Assess the impact for the two data sets separately and the information system as a whole.
e. A power plant contains a SCADA (supervisory control and data acquisition) system controlling the distribution of electric power for a large military installation. The SCADA system contains both real-time sensor data and routine administrative information. Assess the impact for the two data sets separately and the information system as a whole.